

STUDENT: _____ ENTRY POINT: _____

Face sheet

Who she is, and the two things about you that stay the same all term.

PATIENT

Camila Reyes, 19

YOUR
NAME

HISTORY

19 years old.
Second year
kinesiology
student and
a midfielder
on the
college
soccer
team. No
prior medical
history, no
medications.

ALLERGIES

MEDICATIONS AT BASELINE

None

YOUR
ENTRY
POINT,
PICK
ONE IN
WEEK
1 AND
KEEP
IT

Nursing

☐ Medicine

☐ Radiology

☐ Respiratory
therapy

Exercise
☐ and allied
health

—

**BASELINE,
AUGUST 12,
PRESEASON
PHYSICAL
(WEEK 1,
DONE)**

Vitals. HR 52,
BP 108/64, RR
12, temperature
36.8 °C (98.2
°F), SpO₂ 99%
on room air.

Measurements.
Weight 61 kg
(134 lb), height
168 cm (5 ft 6

in). **Labs.**

Sodium 139

mEq/L,

potassium 4.2,

chloride 103,

bicarbonate 25,

glucose 88 mg/

dL, creatinine

0.8 mg/dL,

hemoglobin

13.4 g/dL,

hematocrit

40%. **Note.**

Well. Cleared

for full

participation.

No symptoms,

no medications,

no family

history she

knows of.

STUDENT: _____ ENTRY POINT: _____

Vitals flowsheet weeks 1 to 8

One column per week, in course order. The date under each week is when that chart entry happened in her story; they are not in date order on purpose.

Measure	Wk 1 August 12 Preseason physical	Wk 2 September 1 to 20 Student health visit, September 18	Wk 3 September 22, 06:40 Arrival in the emergency department	Wk 4 September 22, 07:00 to 11:00 Resuscitation, hours 0 to 4	Wk 5 September 22, 06:40 to 14:00 Neurological and autonomic assessment	Wk 6 September 25 to October 10 Mobility assessment and follow-up	Wk 7 September 22 to November 3 Endocrine workup and follow-up	Wk 8 October 20 and December 5 Clinic follow- up, menstrual history
HR	52							
BP	108/64							
RR	12							
Temp °C (°F)	36.8 (98.2)							
SpO ₂	99% RA							
Weight kg (lb)	61 (134)							
Urine output								
Intake								

STUDENT: _____ ENTRY POINT: _____

Vitals flowsheet weeks 9 to 15

Continued.

Measure	Wk 9 September 22 and November 10 Cardiac assessment	Wk 10 September 22, 06:40 to 12:00 Hemodynamics and the response to fluid	Wk 11 September 22 and December 20 Hematology and the three month glycated hemoglobin	Wk 12 October 1 to November 20 Nutrition and energy balance review	Wk 13 September 22, 06:40, and November 25 Ventilation and gas exchange	Wk 14 September 22 to 26, and November 10 Renal function through the admission and after	Wk 15 The whole file Integration
HR							
BP							
RR							
Temp °C (°F)							
SpO ₂							
Weight kg (lb)							
Urine output							
Intake							

STUDENT: _____ ENTRY POINT: _____

Lab flowsheet weeks 1 to 8

Write the units the first time a test appears. A blank row is for a test the week introduces that is not listed.

Measure	Wk 1 August 12 Preseason physical	Wk 2 September 1 to 20 Student health visit, September 18	Wk 3 September 22, 06:40 Arrival in the emergency department	Wk 4 September 22, 07:00 to 11:00 Resuscitation, hours 0 to 4	Wk 5 September 22, 06:40 to 14:00 Neurological and autonomic assessment	Wk 6 September 25 to October 10 Mobility assessment and follow-up	Wk 7 September 22 to November 3 Endocrine workup and follow-up	Wk 8 October 20 and December 5 Clinic follow- up, menstrual history
Sodium mEq/L	139							
Potassium	4.2							
Chloride	103							
Bicarbonate	25							
Glucose mg/dL	88							
BUN mg/dL								
Creatinine mg/dL	0.8							
Osmolality mOsm/kg								
Anion gap								
pH								
PCO ₂ mmHg								
PO ₂ mmHg								
Lactate								
Hemoglobin g/dL	13.4							
Hematocrit %	40							
WBC								
HbA1c %								

STUDENT: _____ ENTRY POINT: _____

Lab flowsheet weeks 9 to 15

Continued.

Measure	Wk 9 September 22 and November 10 Cardiac assessment	Wk 10 September 22, 06:40 to 12:00 Hemodynamics and the response to fluid	Wk 11 September 22 and December 20 Hematology and the three month glycated hemoglobin	Wk 12 October 1 to November 20 Nutrition and energy balance review	Wk 13 September 22, 06:40, and November 25 Ventilation and gas exchange	Wk 14 September 22 to 26, and November 10 Renal function through the admission and after	Wk 15 The whole file Integration
Sodium mEq/L							
Potassium							
Chloride							
Bicarbonate							
Glucose mg/dL							
BUN mg/dL							
Creatinine mg/dL							
Osmolality mOsm/kg							
Anion gap							
pH							
PCO ₂ mmHg							
PO ₂ mmHg							
Lactate							
Hemoglobin g/dL							
Hematocrit %							
WBC							
HbA1c %							

A problem goes on the day you can name it. Write the week you found it, and the week it resolved if it does. Number them; the numbers are how you will refer to them on the week pages.

[illegible]

Fluids, drugs, oxygen, procedures. What was given, when in her story, and what it was meant to change.

[illegible]

STUDENT: _____ ENTRY POINT: _____

02 Student health visit, September 18

September 1 to 20 · The three weeks before she came in · Molecules, Water & Energy

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name the fuel her cells are actually burning and the fuel they cannot get to, and name the single molecule missing between the two.
2. Predict what is happening to her ATP production and to her body mass, separately, and say why those two answers are not the same answer.
3. Explain the mechanism: what insulin normally does at the membrane and inside the cell, and what happens to stored fat and to muscle protein when it is absent.
4. Interpret the dipstick. Say what glucose in the urine tells you, what ketones in the urine tell you, and what neither of them can tell you.
5. Name the one measurement that would separate this from the other things that cause a 19 year old to lose 6 kg in three weeks.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

03 Arrival in the emergency department

September 22, 06:40 · Emergency department, arrival · Membranes, Transport & Compartments

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name the compartments involved and say which solute is setting the gradient between them.
2. Predict which way water has moved between the intracellular and extracellular compartments, and say what has happened to cell volume.
3. Correct her sodium for the glucose. Say which correction factor you used, say that more than one is in use, and say what the corrected number means that the measured one does not.
4. Interpret the measured sodium of 128. Say what it shows and what it cannot show.
5. Name the one measurement that would change your fluid plan in the next hour.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

04 Resuscitation, hours 0 to 4

September 22, 07:00 to 11:00 · The first four hours of treatment · Electrical Signaling

NEW ON THE CHART THIS WEEK

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PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name the ion this entry is about and say which way its electrochemical gradient points across a resting cell membrane.
2. Predict what a serum potassium of 5.4 does to the resting membrane potential, and then what 3.1 does. Give the direction for each.
3. Explain, gate by gate, why a membrane held depolarized becomes less excitable rather than more, and name the channel state that does it.
4. Interpret an arrival potassium of 5.4 in a patient whose total body potassium is depleted by roughly 200 mEq. Say what it shows and what it cannot show.
5. Name the one measurement or event that would make you hold the insulin, and say why holding it is the right move even though the glucose is still high.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

05 Neurological and autonomic assessment

September 22, 06:40 to 14:00 · Arrival through the first afternoon · The Nervous System

NEW ON THE CHART THIS WEEK

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PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name the autonomic outflow that is dominating on arrival, and list three separate findings in this chart that are its signature.
2. Predict what a rapid fall in plasma osmolality would do to brain cell volume, and say why that is the reason for the correction rate in the note.
3. Explain the synapses: name the transmitter and the receptor producing the tachycardia, and the transmitter and receptor producing the cool peripheries. Say why they are different.
4. Interpret reflexes of 1+. Say what that shows and what it cannot show.
5. Name the one measurement that would change how fast you correct her.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

06 Mobility assessment and follow-up

September 25 to October 10 · Hospital day 4 through two weeks after discharge · Muscle

NEW ON THE CHART THIS WEEK

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PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name where in the chain from motor neuron to crossbridge this weakness sits, and say what evidence in the chart rules the other links in or out.

2. Predict what has happened to her maximal force and to her endurance, separately, and say whether they had to change by the same amount.

3. Explain excitation contraction coupling from the action potential to the crossbridge, and say which step is limited when there are simply fewer crossbridges available.

4. Interpret a grip strength of 22 kg against a baseline of 32 kg. Say what it shows and what it cannot show.

5. Name the one measurement that would separate loss of muscle from loss of the ability to activate the muscle she still has.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

07 Endocrine workup and follow-up

September 22 to November 3 · Admission through the six week clinic visit · Chemical Signaling & Endocrine Control

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name each hormone in this entry, its main target tissue, and the type of receptor it uses.
2. Predict which way glucagon, cortisol, growth hormone and the catecholamines each moved on arrival, and say why they all moved in the same direction at the same time.
3. Explain what C-peptide is, where it comes from, and why it separates her from a patient with type 2 diabetes who is taking insulin.
4. Interpret the thyroid results across the three dates. Say what they show and what they cannot show, and say whether her thyroid was ever the problem.
5. Name the one measurement that would change her insulin plan in the next month.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

08 Clinic follow-up, menstrual history

October 20 and December 5 · Four weeks and eleven weeks after discharge · Reproductive Physiology

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name the three levels of this axis and the hormone each one releases.
2. Predict which way FSH, LH and estradiol each moved, and say what that particular combination localizes.
3. Explain what GnRH pulse frequency has to do with which gonadotropin the pituitary favors, and what slows that frequency.
4. Interpret low FSH with low estradiol. Say what it shows and what it cannot show, then contrast it with high FSH and low estradiol.
5. Name the one measurement that would tell you the axis was recovering before her period returned.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

09 Cardiac assessment

September 22 and November 10 · Arrival, and the seven week echocardiogram · Cardiac Function

NEW ON THE CHART THIS WEEK

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PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name which term of cardiac output is doing the work on arrival, and say what the other term is doing.
2. Predict her stroke volume on arrival relative to her August baseline, and justify the prediction from the echocardiogram rather than from the blood pressure.
3. Explain, through the cardiac cycle, why the small end diastolic volume and the rate of 128 are the same problem and not two problems.
4. Interpret an ejection fraction of 62% in a patient with a low cardiac output. Say what it shows and what it cannot show.
5. Name the one measurement that would tell you whether more fluid will raise her cardiac output.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

10 Hemodynamics and the response to fluid

September 22, 06:40 to 12:00 · The first five hours · Circulation & Blood Pressure

NEW ON THE CHART THIS WEEK

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PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name what is being defended here and name what is being spent in order to defend it.

2. Predict what cardiac output and total peripheral resistance were each doing at 06:40, and say how a nearly normal blood pressure is consistent with both answers.

3. Explain the baroreflex from receptor to effector: where the receptors are, what they actually sense, and what each efferent limb does.

4. Interpret a supine blood pressure of 96/58 in this patient. Say what it shows and what it cannot show.

5. Name the one measurement that would tell you the reflex has stopped compensating.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

11 Hematology and the three month glycated hemoglobin

September 22 and December 20 · Arrival, and the three month review · Blood & Immunity

NEW ON THE CHART THIS WEEK

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PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name exactly what hematocrit is a measurement of, including what sits in the denominator.
2. Predict what her hematocrit will do over the first twelve hours of rehydration if not a single red cell is added or lost, and say why.
3. Explain what HbA1c measures and why the lifespan of a red cell sets the window it reports on.
4. Interpret a white cell count of 18.4 with no fever and negative cultures. Say what it shows and what it cannot show.
5. Name the one measurement that would tell you whether her red cell mass was ever abnormal at all.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

12 Nutrition and energy balance review

October 1 to November 20 · Two weeks to eight weeks after discharge · Digestion, Absorption & Energy Balance

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name the step in the digestive sequence that early fullness points to, and say what normally happens at that step.
2. Predict what happens to her weight if intake exceeds expenditure by roughly 300 kcal a day for seven weeks, and say what that regained weight is made of.
3. Explain what indirect calorimetry actually measures at the mouth, and how those measurements become a figure in kilocalories per day.
4. Interpret a measured resting energy expenditure below the predicted value. Say what it shows and what it cannot show.
5. Name the one measurement that would separate slow gastric emptying from a stomach that has simply become small.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

13 Ventilation and gas exchange

September 22, 06:40, and November 25 · Arrival, and the return to training test · Respiratory Physiology

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name whether the problem on arrival is ventilation, gas exchange, or gas transport, and justify the answer from the numbers rather than from the diagnosis.
2. Predict roughly what her alveolar ventilation must be, given a PaCO_2 of 14 mmHg, and say what that is as a multiple of normal.
3. Explain the relationship between alveolar ventilation and PaCO_2 , and say what it predicts if ventilation is doubled and then doubled again.
4. Interpret an oxygen saturation of 99% in a patient this sick. Say what it shows and what it cannot show.
5. Name the one measurement that would tell you her respiratory compensation is beginning to fail, and say what you would do about it.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

14 Renal function through the admission and after

September 22 to 26, and November 10 · The admission, and the seven week review · Renal Physiology & Fluid Balance

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Name what her kidney is defending on September 22, and name what it is spending in order to defend it.
2. Predict what happened to GFR, to urine volume and to urine sodium, separately, and explain how urine volume and GFR moved in opposite directions.
3. Explain the transport maximum for glucose and how exceeding it produces an osmotic diuresis.
4. Interpret a urine sodium of 14 mEq/L alongside a creatinine of 1.6. Say what it shows and what it cannot show.
5. Name the one measurement that would tell you this was intrinsic kidney injury rather than a kidney responding correctly to a low circulating volume.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

15 Integration

The whole file · Reading September 22 again, with everything you now know · Acid-Base Balance & Integration

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Use the numbers from your problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

1. Read the pH on arrival and name the primary disturbance. Then do the same for the 18:00 gas.
2. Predict what the PaCO_2 should be at each time point using Winter's formula, and say whether the compensation is appropriate at each.
3. Calculate the anion gap on arrival and at 18:00, and explain why the gap closed while the chloride rose by 16 mEq/L.
4. Interpret the 18:00 gas. Say what it shows and what it cannot show about whether she is getting better.
5. Name the one measurement that would change your management at 18:00, and say what you would do differently.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.